

Chemical Names and Formulas

Answer key with worked solutions

A. Ionic or molecular?

1. Classify.

(a) **ionic** (metal K) (b) **molecular** (non-metals only) (c) **ionic** (ammonium ion) (d) **molecular** (e) **ionic** (metal Fe)

B. Ionic compounds

2. Formulas.

- (a) K^+ , O^{2-} : $2(+1) + (-2) = 0$, so K_2O
(b) Mg^{2+} , N^{3-} : $3(+2) + 2(-3) = 0$, so Mg_3N_2
(c) Zn^{2+} , Cl^- : $(+2) + 2(-1) = 0$, so $ZnCl_2$
(d) Ba^{2+} , SO_4^{2-} : $(+2) + (-2) = 0$, so $BaSO_4$

3. Formulas with polyatomic ions.

- (a) Ca^{2+} , NO_3^- : $(+2) + 2(-1) = 0$, so $Ca(NO_3)_2$
(b) Al^{3+} , OH^- : $(+3) + 3(-1) = 0$, so $Al(OH)_3$
(c) NH_4^+ , CO_3^{2-} : $2(+1) + (-2) = 0$, so $(NH_4)_2CO_3$
(d) Na^+ , HCO_3^- : $(+1) + (-1) = 0$, so $NaHCO_3$

4. Names.

(a) **sodium sulfide** (b) **sodium sulfate** (c) **silver nitrate** (d) **magnesium phosphate**

5. Roman numerals.

- (a) 2 Cl^- give -2 , so Fe is $+2$: **iron(II) chloride**
(b) 3 O^{2-} give -6 , shared by 2 Fe, so each Fe is $+3$: **iron(III) oxide**
(c) 1 O^{2-} gives -2 , shared by 2 Cu, so each Cu is $+1$: **copper(I) oxide**
(d) 1 SO_4^{2-} gives -2 , so Cu is $+2$: **copper(II) sulfate**

C. Molecular compounds

6. Prefixes.

(a) **carbon dioxide** (b) **phosphorus pentachloride** (c) **dinitrogen monoxide** (nitrous oxide) (d) **sulfur hexafluoride** (e) SO_3 (f) N_2O_4

D. Acids

7. Acids.

(a) HCl (b) HNO_3 (c) H_2SO_4 (d) H_3PO_4

E. Challenge

8. Spot the errors.

- Copper forms Cu^+ and Cu^{2+} , so the name needs a Roman numeral. In copper(II) sulfate, $(+2) + (-2) = 0$, so one sulfate is enough: **copper(II) sulfate is CuSO_4** . $\text{Cu}(\text{SO}_4)_2$ would need copper to be +4.
- CaCl_2 is ionic, so prefixes are not used, and the formula contains one calcium, not two: **calcium chloride**.